TV programme recommendation

Midterm | Gleb Alektorov | 24 October 2026

Predicting five preferred programmes from viewing intervals and the broadcast schedule. Implementation, chronological evaluation and platform strategy.

Repository: https://github.com/Moonkato/llm4rec_midterm/tree/main/solutions/midterm

The implementation and final CSV are packaged for this repository; remote publication is performed by the repository owner. Code, evidence and candidate CSVs are stored in solutions/midterm.

1. Task and data

The output is exactly five ordered, distinct, nonzero tv_show_id values for each of 1,260 template users. The forecast covers weeks 31-42 (27 July to 19 October 2020). MAP@5 rewards relevant programmes near the top of the list. User IDs are preserved as strings to avoid 64-bit precision loss.

The supplied data contains 5,821,987 viewing intervals and three EPG schedule files: 1,074,235, 1,074,394 and 1,295,378 rows. A viewing duration is session length, not programme completion. The median raw session lasts 396 seconds, while extreme sessions can span days. Only 10.8% of schedule rows in weeks 11-20 carry a programme ID; this percentage describes rows, not duration-weighted airtime coverage.

A session [s,e) is matched to an airing [a,b) only on the same channel, with overlap max(0, min(e,b)-max(s,a)). Airing duration D=b-a provides the denominator. All historical features stop at the relevant cutoff. Interval timestamps use the repository clock conversion, including its March 29 one-hour adjustment; replay checks cover missing-hour and midnight cases.

Initially, an airing was watched if merged overlapping fragments covered at least 180 seconds. This filters brief channel switches relative to the observed session median, but does not account for programme length. The later completion-oriented protocol uses more than 30% of an airing for history and more than 80% for target counts, using a single overlapping fragment rather than the union of interrupted fragments. The legacy target takes the maximum stored overlap fragment; fragments separated by the artificial May 18 preprocessing boundary are not rejoined. Replay features instead use raw continuous sessions. This is an operational approximation to preferred programmes, not a verified reproduction of the hidden Kaggle label generator.

ID 0 is excluded from every item list and target. Its sessions can still describe channel/time viewing context. Future EPG supplies programme availability, channel, start time and duration only; no test-period views enter training or features. Metadata for the content model comes from the historical scope.

2. Evaluation contract

The main holdout is 18 May-27 July 2020; histories end on 18 May. The later robustness window is 22 June-27 July, with history ending on 22 June. These windows overlap and are not independent test sets. Supervised rankers use an earlier fit task: history 9 March-13 April and labels 13 April-18 May. Repeated validation-guided experimentation creates selection optimism; the later window reduces but does not eliminate it.

For user u, AP@5 = sum over ranks r=1..5 of Precision@r times relevance(r), divided by min(5, number of relevant items). Users with empty truth receive zero. MAP@5 averages all 1,260 users. The initial target is all distinct qualifying 180-second items; the later target selects the five most frequent qualifying >80% airings, with numeric-ID tie breaking. These are different metrics even though both are called MAP@5.

3. Implemented methods and common output rules

Every system permits repeats of previously watched programmes and restricts output to nonzero programmes airing during the forecast window. Cold users receive historical unique-viewer popularity restricted to eligible items. Zero-signal or short lists are completed by that common fallback without duplicates. Ties use ascending numeric programme ID; final rankings are truncated to five. The fallback rule is shared, but the number of cold users can change with the historical qualification threshold.

Content-based filtering

Historical item attributes are tokenised into category, genre, director, actor and year/decade blocks. IDF is log((1+N)/(1+df))+1. Each block is L2-normalised and weighted by 1.0, 1.5, 0.75, 1.0 and 0.35 respectively; the concatenated item vector is normalised again. A user profile is the normalised sum of item vectors weighted by log(1+historical airing count). Cosine similarity produces a ranking, not a viewing probability. Missing metadata gives a zero vector; there are no title or identity tokens. The observed vocabulary has 11,939 features across the five blocks.

Implicit-feedback matrix factorization

ALS minimises sum c_ui(p_ui-x_u dot y_i)^2 plus lambda times the squared factor norms, with p_ui=1 for a positive count and c_ui=1+alpha*log(1+count_ui). Missing pairs have preference zero and confidence one; they are not treated as explicit dislikes. Alternating normal equations are solved in float64, and stored factors use float32. Dot products rank eligible programmes.

The historical matrix contains 4,851 users, 8,233 items and 651,903 nonzeros. Tested (factors, alpha) pairs were (32,1), (32,10), (64,10), (64,40); regularization=1, iterations=10, seed=20261022. The best initial ALS variant was 32 factors, alpha=1. No validation labels enter factor fitting.

Popularity and personal frequency

Popularity ranks by distinct historical viewers. Personal frequency ranks each user's historical qualifying airing counts and uses popularity to fill zero-signal slots. Popularity is a required baseline and personal frequency is an additional heuristic; neither is counted as one of the two required methods. Content filtering and implicit ALS satisfy that requirement.

Initial controlled comparison

All rows below use the same 180-second histories, holdout, all-item truth, eligible catalogue, repeat policy and fallback. Fit times exclude data preprocessing. These high all-item scores must not be interpreted as competition scores.

popularity_unique_viewers: MAP@5 0.752995; fit 0.000 s; ranking/evaluation 0.201 s.

personal_frequency: MAP@5 0.776757; fit 0.000 s; ranking/evaluation 0.072 s.

content_tfidf: MAP@5 0.064574; fit 0.320 s; ranking/evaluation 0.334 s.

als_f32_a1: MAP@5 0.598164; fit 2.622 s; ranking/evaluation 0.313 s.

The complete initial round took 23.22 seconds and peaked at 385 MB RSS on macOS arm64, Python 3.13.1, NumPy 2.5.4 and SciPy 1.18.1, using one compute thread. ALS variants with larger confidence or dimension performed worse here; increasing capacity was not evidence of better generalization.

4. Schedule-aware ranking and final strategy

The weekly ranker v3 combines 19 history/popularity/channel/hour/availability features with five weekly-context features: expected minutes, expected airing fraction, recent-28-day minutes, activity frequency and a recent/all-history ratio. Candidate retrieval combines personal candidates (up to 100), popularity (30) and channel-hour candidates (30), then deduplicates. This shortlist is a system component, not the entire eligible EPG catalogue.

LightGBM LambdaRank uses 417 trees, 15 leaves, maximum depth 4, minimum child count 100, L2=10, learning rate 0.03, label gains [0,1], truncation 10 and seed 20261023. The configuration and 417-tree count were selected from three configurations on earlier inner-fit/inner-stop temporal tasks with early-stopping patience 100, then refitted on the April 13-May 18 task. Training retains queries with both relevant and nonrelevant candidates. It learns ordering against the top-five completion-count target. All fitted feature transforms and the model are frozen before each evaluation forecast.

The airing model scores individual candidate airings from 33 features: the history/context base plus channel, weekday, hour, duration, availability and session-coverage features. A binary LightGBM with 300 trees, 15 leaves, learning rate 0.05, minimum leaf count 100, L2=10 and max_bin=63 and seed=20261023 is trained on 473,908 sampled rows (102,907 positive; 371,001 negative). Negative sampling probability is 0.01, with inverse-sampling weights 100 for negatives and 1 for positives. Summed per-airing predictions form an expected-count ranking; they are not calibrated satisfaction probabilities.

Hybrid v4 combines v3 and the airing model with equal reciprocal-rank fusion: score(i)=0.5/(60+rank_v3(i))+0.5/(60+rank_event(i)). Rank fusion avoids pretending that a LambdaRank score and an expected count share a probability scale. Equal weights and the RRF constant 60 were prespecified in the airing-model protocol, rather than optimized on the final holdout. This fixed fusion improved both chronological windows relative to either component. It was the strongest candidate among the first four submissions; the fifth submission subsequently improved both returned scores.

Matched-pool system comparison

All ten frozen methods are evaluated on exactly the same per-user shortlist (30-151 candidates; 120,172 rows total), the same 1,260 users and the same full-universe top5_count80 truth. Repeats are allowed. A common 12-user cold cohort is the union of users without >=180-second history or without >30%-coverage history. Every method returns the identical historical popularity fallback [2400480, 240081, 2400508, 6500479, 2400467]; these five IDs occur in every shortlist. Training-label definitions still differ. This audit performs no fitting and does not replace the submitted CSV.

popularity_unique_viewers: strict top-five-count MAP@5 = 0.270649.

content_tfidf: strict top-five-count MAP@5 = 0.013115.

als_f32_a1: strict top-five-count MAP@5 = 0.160028.

personal_frequency: strict top-five-count MAP@5 = 0.339491.

weekly_ranker_v3: strict top-five-count MAP@5 = 0.407363.

airing_model: strict top-five-count MAP@5 = 0.412203.

hybrid_v4: strict top-five-count MAP@5 = 0.417187.

identity_replay_v5: strict top-five-count MAP@5 = 0.419172.

identity_ranker: strict top-five-count MAP@5 = 0.412695.

replay_airing: strict top-five-count MAP@5 = 0.413059.

These matched-pool results supersede the earlier mixed-retrieval table for comparing systems. The original unmodified v4 validation remains 0.417238 (main) and 0.365450 (later); the small main-window change here is solely the standardized fallback. No Kaggle score is attributed to this audit wrapper. The initial 180-second table remains the controlled model-family comparison with identical historical labels.

Robustness window: v3 0.361875; airing 0.356944; v4 0.365450. Main-window v4 improves on v3 by 0.009825 and on the airing model by 0.004984. Reusing the same holdout for many experiments limits the certainty of these gains.

5. Final experiment, submission evidence and limitations

The final experiment added categorical user/programme IDs to the original 24-feature ranker (26 features), and exact historical same-weekday/channel session-replay features to the original 33-feature airing model (37 features). Encoders were fitted only on training rows; unknown categories receive -1. Replay features measure >30% and >80% single-session overlap, a recent-28-day completion rate and observable historical-date count. Sessions are clipped at the cutoff; interrupted fragments are never merged into a synthetic continuous session.

Two fits completed in 791.59 seconds with 1.026 GB peak RSS. The combined identity/replay arm reached 0.419222 on the main holdout and 0.364225 on the later window, versus v4 at 0.417238 and 0.365450. Main delta +0.001984 has paired 95% interval [-0.001883, 0.005796]; later delta -0.001225 has interval [-0.005427, 0.002919]. It failed the prespecified consistency screen. The combined CSV was submitted for the final external check. Kaggle returned Private 0.44559 and Public 0.45546, both higher than v4. The screenshot below confirms the outcome; it does not erase the earlier mixed validation evidence.

Earlier attempts included expanded weekly candidates, EASE features, rolling and full-history training, recurrence features, alternative objectives and user-fold out-of-fold stacking. None established a clear, consistent replacement for v4. LightFM installation failed under Python 3.13 and no LightFM training result is claimed. No external or synthetic viewing records were added. Stopping avoids spending more compute on small, uncertain offline gains.

Kaggle submission record: Private / Public

01 logistic_v1.csv: 0.42693 / 0.44458

02 filtered_context_v2.csv: 0.42939 / 0.44481

03 weekly_ranker_v3.csv: 0.43519 / 0.44966

04 event_v3_rrf_v4.csv: 0.44394 / 0.45348

05 final_factorial_v5.csv: 0.44559 / 0.45546 (designated final submission)

The supplied Kaggle screenshot confirms the Private/Public column order and the v5, v4 and v3 results. The v2 row is partially visible and agrees with the earlier reported pair; v1 remains an account-owner report. The designated final file is candidates/final_factorial_v5.csv, description gleb_alektorov_midterm_05_identity_replay_rrf, SHA-256 662620aafd8df3aaf1230f7653c0910708d5fe8fbc056b28b78bb8ddd34a88a3. It improves on v4 by 0.00165 Private and 0.00198 Public. It contains 1,260 ordered users and five unique eligible IDs per user.

Submission evidence

The screenshot contains the actual filename, description, completion status and both score-column headings. Thus it supplies the score evidence requested by either version of the assignment: the original PDF specifies Private score, while the Markdown specifies Public score. The repository owner will publish the implementation and final CSV; remote publication is not yet verified.

Target ambiguity matters: frequency ties affect roughly 58-61% of users. Exact tie-averaged AP gives v4 0.393055 / 0.335472; a numerical-ID tie rule is a local convention. Candidate oracles and perfect reranking figures are upper bounds computed with holdout knowledge, not deployable predictions. The observed data provides no evidence that 0.8 is attainable or that the current solution is optimal.

Kaggle evidence: final submission

Final: final_factorial_v5.csv | Private MAP@5 0.44559 | Public MAP@5 0.45546

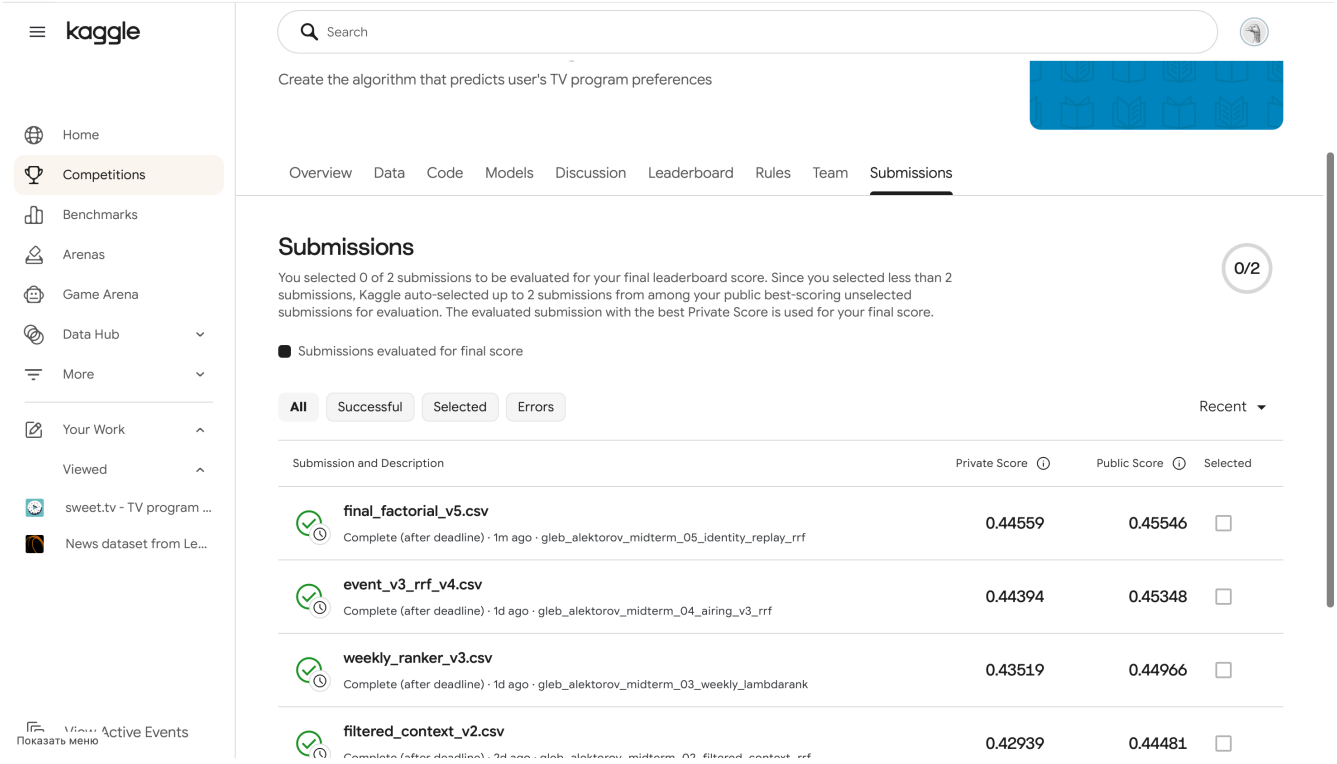


Figure 1. Original screenshot supplied by the account owner. The first row identifies the final identity/replay RRF submission; the following rows show earlier candidates. Both score headings are visible. Image is embedded without altering its contents.

Final fusion: equal RRF with constant 60 over the 26-feature identity ranker and the 37-feature replay airing model. Original main/later MAP@5: identity alone 0.412746 / 0.361151; replay alone 0.413109 / 0.353308; combined 0.419222 / 0.364225. Thus the combined model outperforms either new component individually on both local windows, while still trailing the older v4 on the later window. The shared-pool table and user examples also include both standalone components.

6.1. User example: 15634861151839696048

Selection: 25th percentile of non-cold historical airing count; 199 historical qualifying airings and 92 distinct programmes. Users were selected by activity quartile rather than by model success. The same two users are evaluated across all ten methods under the matched-pool protocol. The lists below are the corrected matched-pool outputs; original full-catalogue outputs remain in the evidence.

Actual strict holdout top five

1. 240081 | Факты. День. | Инфо
2. 2400480 | 'ТСН. Неделя'. | Инфо
3. 20088 | 'Завтрак с 1+1'. Информационно-развлекательная программа. | Развлечения
4. 1100411 | Премьера. 'Вещдок. Особый случай. Око за око'. | Познавательное
5. 12001697 | М/с 'Аватар: Легенда об Аанге', 56 с. 'Южные захватчики'. | Детям

Programme titles/categories are representative EPG labels for an ID and may contain episode-specific wording. H = a hit in the strict top-five target; S = present in the historical 180-second history. An absent H does not mean no viewing at all, and an absent S does not establish that the person has never seen the programme.

popularity_unique_viewers | AP@5 0.4000

1. 2400480 | 'ТСН. Неделя'. | Инфо | H=1 S=1
2. 240081 | Факты. День. | Инфо | H=1 S=1
3. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1
4. 6500479 | 'Мир наизнанку. Бразилия', 36 эп. | Познавательное | H=0 S=0
5. 2400467 | 'Викна-Новости'. | Инфо | H=0 S=1

content_tfidf | AP@5 0.0000

1. 90079212 | Х/ф 'Боги Египта'. | Фильмы | H=0 S=1
2. 90057593 | Х/ф 'Сокровище Амазонки'. | Фильмы | H=0 S=1
3. 90071644 | Х/ф 'Принц Персии: Пески времени'. | Фильмы | H=0 S=1
4. 90067713 | Х/ф 'Хеллбой 2. Золотая армия'. | Фильмы | H=0 S=1
5. 90081867 | Премьера. Х/ф 'Миссия невыполнима 6. Последствия'. | Фильмы | H=0 S=1

als_f32_a1 | AP@5 0.5500

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=1 S=1
3. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1
4. 20088 | 'Завтрак с 1+1'. Информационно-развлекательная программа. | Развлечения | H=1 S=1
5. 2500413 | Гражданская оборона. | Инфо | H=0 S=1

personal_frequency | AP@5 0.4533

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 1000723 | Т/с 'Охотники за древностями', 3 сезон, 14 и 15 с. | Сериалы | H=0 S=1
3. 2400480 | 'ТСН. Неделя'. | Инфо | H=1 S=1
4. 200469 | Варьяты. Лучшее. | Развлечения | H=0 S=1
5. 1100411 | Премьера. 'Вещдок. Особый случай. Око за око'. | Познавательное | H=1 S=1

6.1. User 15634861151839696048: schedule-aware methods

weekly_ranker_v3 | AP@5 0.7600

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=1 S=1
3. 20088 | 'Завтрак с 1+1'. Информационно-развлекательная программа. | Развлечения | H=1 S=1
4. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1
5. 1100411 | Премьера. 'Вещдок. Особый случай. Око за око'. | Познавательное | H=1 S=1

airing_model | AP@5 0.8000

1. 2400480 | 'ТСН. Неделя'. | Инфо | H=1 S=1
2. 240081 | Факты. День. | Инфо | H=1 S=1
3. 20088 | 'Завтрак с 1+1'. Информационно-развлекательная программа. | Развлечения | H=1 S=1
4. 1100411 | Премьера. 'Вещдок. Особый случай. Око за око'. | Познавательное | H=1 S=1
5. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1

hybrid_v4 | AP@5 0.8000

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=1 S=1
3. 20088 | 'Завтрак с 1+1'. Информационно-развлекательная программа. | Развлечения | H=1 S=1
4. 1100411 | Премьера. 'Вещдок. Особый случай. Око за око'. | Познавательное | H=1 S=1
5. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1

identity_replay_v5 | AP@5 0.8000

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=1 S=1
3. 20088 | 'Завтрак с 1+1'. Информационно-развлекательная программа. | Развлечения | H=1 S=1
4. 1100411 | Премьера. 'Вещдок. Особый случай. Око за око'. | Познавательное | H=1 S=1
5. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1

identity_ranker | AP@5 0.7600

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=1 S=1
3. 20088 | 'Завтрак с 1+1'. Информационно-развлекательная программа. | Развлечения | H=1 S=1
4. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1
5. 1100411 | Премьера. 'Вещдок. Особый случай. Око за око'. | Познавательное | H=1 S=1

replay_airing | AP@5 0.8000

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=1 S=1
3. 20088 | 'Завтрак с 1+1'. Информационно-развлекательная программа. | Развлечения | H=1 S=1
4. 1100411 | Премьера. 'Вещдок. Особый случай. Око за око'. | Познавательное | H=1 S=1
5. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1

6.2. User example: 10269329346053634491

Selection: 75th percentile of non-cold historical airing count; 503 historical qualifying airings and 251 distinct programmes. Users were selected by activity quartile rather than by model success. The same two users are evaluated across all ten methods under the matched-pool protocol. The lists below are the corrected matched-pool outputs; original full-catalogue outputs remain in the evidence.

Actual strict holdout top five

1. 1100412 | 'Вещественное доказательство'. | Познавательное
2. 10003167 | Т/с 'Крепостная', 1 и 2 с. | Сериалы
3. 90072814 | Х/ф 'Исходный код'. | Фильмы
4. 240081 | Факты. День. | Инфо
5. 2400467 | 'Викна-Новости'. | Инфо

Programme titles/categories are representative EPG labels for an ID and may contain episode-specific wording. H = a hit in the strict top-five target; S = present in the historical 180-second history. An absent H does not mean no viewing at all, and an absent S does not establish that the person has never seen the programme.

popularity_unique_viewers | AP@5 0.1800

1. 2400480 | 'ТСН. Неделя'. | Инфо | H=0 S=1
2. 240081 | Факты. День. | Инфо | H=1 S=1
3. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1
4. 6500479 | 'Мир наизнанку. Бразилия', 36 эп. | Познавательное | H=0 S=1
5. 2400467 | 'Викна-Новости'. | Инфо | H=1 S=1

content_tfidf | AP@5 0.0000

1. 90074614 | Х/ф 'Притворись моим мужем'. | Фильмы | H=0 S=1
2. 90076668 | Х/ф 'Два ствола'. | Фильмы | H=0 S=1
3. 90028050 | Х/ф 'Жандарм из Сен-Тропе', 1 ч. | Фильмы | H=0 S=1
4. 90055201 | Х/ф 'Интимный словарь', 1 ч. | Фильмы | H=0 S=1
5. 90046977 | Х/ф 'Любимая женщина механика Гаврилова'. | Фильмы | H=0 S=1

als_f32_a1 | AP@5 0.0000

1. 300358 | Скетч-шоу 'На троих', 7 сезон. | Развлечения | H=0 S=1
2. 200337 | 'Утро с 'Интером'. | Развлечения | H=0 S=1
3. 2500430 | 'Затерянный мир'. | Познавательное | H=0 S=1
4. 20088 | 'Завтрак с 1+1'. Информационно-развлекательная программа. | Развлечения | H=0 S=1
5. 200405 | 'Мамахохотала-шоу'. Лучшее. | Развлечения | H=0 S=1

personal_frequency | AP@5 0.1000

1. 2400480 | 'ТСН. Неделя'. | Инфо | H=0 S=1
2. 240081 | Факты. День. | Инфо | H=1 S=1
3. 2500413 | Гражданская оборона. | Инфо | H=0 S=1
4. 5200303 | 'Следствие вели...' с Леонидом Каневским. | Познавательное | H=0 S=1
5. 200405 | 'Мамахохотала-шоу'. Лучшее. | Развлечения | H=0 S=1

6.2. User 10269329346053634491: schedule-aware methods

weekly_ranker_v3 | AP@5 0.2000

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=0 S=1
3. 200432 | Дизель-шоу. 8 марта. | Развлечения | H=0 S=1
4. 2500413 | Гражданская оборона. | Инфо | H=0 S=1
5. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1

airing_model | AP@5 0.2000

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=0 S=1
3. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1
4. 10003012 | Т/с 'Однажды в Одессе'. | Сериалы | H=0 S=1
5. 1100411 | Премьера. 'Вещдок. Особый случай. Око за око'. | Познавательное | H=0 S=1

hybrid_v4 | AP@5 0.2000

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=0 S=1
3. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1
4. 2500413 | Гражданская оборона. | Инфо | H=0 S=1
5. 10003012 | Т/с 'Однажды в Одессе'. | Сериалы | H=0 S=1

identity_replay_v5 | AP@5 0.2000

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=0 S=1
3. 2500413 | Гражданская оборона. | Инфо | H=0 S=1
4. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1
5. 10003012 | Т/с 'Однажды в Одессе'. | Сериалы | H=0 S=1

identity_ranker | AP@5 0.1000

1. 2400480 | 'ТСН. Неделя'. | Инфо | H=0 S=1
2. 240081 | Факты. День. | Инфо | H=1 S=1
3. 2500413 | Гражданская оборона. | Инфо | H=0 S=1
4. 200432 | Дизель-шоу. 8 марта. | Развлечения | H=0 S=1
5. 200405 | 'Мамахохотала-шоу'. Лучшее. | Развлечения | H=0 S=1

replay_airing | AP@5 0.2000

1. 240081 | Факты. День. | Инфо | H=1 S=1
2. 2400480 | 'ТСН. Неделя'. | Инфо | H=0 S=1
3. 2400508 | Сегодня. Итоги с Олегом Панютой. | Инфо | H=0 S=1
4. 10003012 | Т/с 'Однажды в Одессе'. | Сериалы | H=0 S=1
5. 2500413 | Гражданская оборона. | Инфо | H=0 S=1

7. Platform choice and interpretation

I designate v5 (identity ranker plus replay airing model, equal RRF60) as the final competition strategy because the completed external check improved both Private and Public MAP@5 over v4. V4 remains the stronger reference on the later local window. For a platform, v5 is the candidate I would take forward to an online evaluation, with popularity fallback retained; the small offline gains and mixed local evidence do not establish an online benefit or statistical superiority on hidden labels.

For user 15634861151839696048, the historical profile includes news programmes such as Facts and TSN. Popularity achieves AP@5=0.40, ALS 0.55, the weekly ranker 0.76 and v4 0.80. The schedule-aware system largely recovers recurring preferences. This is useful familiarity, not evidence of discovery or enjoyment. Content filtering scores zero despite descriptive attributes: semantic resemblance alone does not recover this user's five most frequent future programmes.

For user 10269329346053634491, more history does not make the prediction easy. Popularity reaches 0.18, content and ALS zero, personal frequency 0.10, and v4 0.20. The actual top five contain programmes missed by the models. The contrast supports a failure case involving changing or insufficiently represented preferences; it does not establish that all heavy viewers are difficult. The complete ordered IDs and observed hits above make this judgment inspectable.

ALS shares statistical strength through low-rank co-preference patterns, but this task rewards repeated schedules and strongly time-dependent availability. Content features can connect related metadata where co-view data is thin, yet missing or generic attributes and absent historical vectors for new items limit that benefit here. Association rules would express direct conditional co-occurrence; graph diffusion would spread evidence through multiple co-view links. Those mechanisms differ from ALS, but no unreported association/PPR score is used to justify the final choice.

Popularity already captures broadly watched news and is extremely cheap. It cannot express the user's programme-specific weekday routine. The hybrid costs more, particularly when evaluating millions of airings, but runs at this scale within about 1 GB and minutes rather than requiring a GPU. Candidate caching and collapsed repeated airing patterns reduce inference cost. V5 export took 399.66 seconds, peaked at 838 MB and scored 8,297,845 replay tuples without new fits.

Coverage is the primary limitation: most EPG rows lack item IDs, so both learning and evaluation ignore much of what people watch. Non-viewed programmes are not proven dislikes, long sessions may be passive background viewing, and device/channel activity does not reveal enjoyment. Completion counts overweight repeated short programmes and changing broadcast supply. Only three test users are cold under the final export protocol, but sparse-history and new-programme weaknesses remain.

No live A/B test, revenue lift, retention effect or causal business outcome was measured. An operational decision should preserve repeat recommendations where supported by behaviour and evaluate discovery separately; unfamiliar output is not automatically a better recommendation. The available evidence supports choosing the hybrid for this dataset, not a universal ranking of recommendation families.

8. Verification and reproducibility

The final audit reran 16 independent hand-calculated AP@5 fixtures: all passed. Re-evaluation of stored scores reproduced the reported original metrics. Both v4 and v5 CSVs were independently checked against the template, eligible catalogue and saved fusion scores: all 1,260 ranked lists match, with five distinct nonzero IDs each. Source review found no future-viewing feature leakage in the examined builders. This was a source/artifact audit, not a fresh full raw-data rebuild or proof that every possible defect is absent.

A shared-candidate/shared-fallback audit closed the comparison discrepancy between earlier baselines and advanced systems. The expanded ten-method audit took 3.02 seconds without fitting. Reproduction: solutions/midterm/.venv/bin/python solutions/midterm/tools/build_matched_pool_comparison.py. Evidence: evidence/matched_pool_comparison.json, audit_code_final.json and audit_session_final.json.

Implemented sources: models/content.py and models/implicit_als.py; tools/build_interactions.py and run_round1.py; tools/run_deep_temporal.py, run_airing_model.py and export_airing_fusion.py; tools/run_final_identity.py, run_final_replay.py and export_final_factorial.py. Evidence is stored in per-experiment JSON files with protocols, metrics, runtimes and hashes. tools/build_report_comparison.py re-evaluates saved predictions without training. Tools/build_report.py and render_report.swift regenerate this document from evidence.

Run from the repository root using solutions/midterm/.venv/bin/python. The saved environment is Python 3.13 on Apple arm64 (Python 3.13.1, NumPy 2.5.4, SciPy 1.18.1, LightGBM 4.7.0); seeds and thread limits are fixed in the relevant scripts. Raw data, caches and local checkpoints are required for full reproduction and are not implied to be committed. No fresh full end-to-end rerun was performed while writing this report.

The supplied competition data is the empirical source; no external viewing dataset or pretrained recommender was used. External methodological guidance included LightGBM categorical-feature documentation and the published two-layer cinema recommender solution. Its reported score was not treated as a target guarantee.

References

Assignment: midterm/LLM4Rec. Midterm Quiz.pdf and midterm/exam_quiz_GSB.md (sections 1-10); score-column requirements conflict.

Competition: https://www.kaggle.com/competitions/sweettv-tv-program-recommender

LightGBM: https://lightgbm.readthedocs.io/en/latest/Advanced-Topics.html

Nikita Melentev, Two-layered Recommender System Methodology: A Prize-Winning Solution, 20 March 2021: https://medium.com/data-science/two-layered-recommender-system-methodology-a-prize-winning-solution-c4bdb815b156

Primary local evidence: evidence/report_comparison.json; evidence/round1/protocol.json; evidence/airing_model/results.json; evidence/final_factorial/results.json; evidence/submissions/; evidence/candidate_final_factorial_v5/manifest.json.